

Beyond Human Bonds: Understanding Relationship Development With Humanized AI Companions

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Abstract

This study investigates the development of relationships between humans and Humanized AI Companions (HACs), focusing on the chatbot Replika. Drawing on interpersonal relationship theories, we explore how these relationships evolve, emphasizing the role of self-disclosure and communication. Through 12 in-depth interviews with Replika users we reveal that users perceive HACs as emotionally supportive, nonjudgmental companions, fostering meaningful bonds despite the AI's lack of true reciprocity. Relationship development follows stages similar to human relationships, with self-disclosure playing a key role in deepening intimacy. However, the relationships are often asymmetrical, with users deriving emotional benefits without expecting mutual exchange. Technical limitations, such as memory issues, are acknowledged but do not significantly hinder relationship progression. Our study challenges traditional interpersonal relationship theories, suggesting that human-HAC relationships are hybrid in nature, blending human-like and machine-like characteristics.

Keywords: chatbot, companionship, relational hybridity, self-disclosure

Introduction

The emergence of AI chatbots has introduced a new frontier for communication studies that challenges traditional models centered on human-to-human interaction. Communication, long regarded as a uniquely human activity, is now being reshaped by the

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interactions between humans and AI (Greussing et al., 2022; Hepp, 2020; Lewis et al., 2019). One significant development in this realm is the rise of Humanized AI Companions (HACs), AI-driven entities designed to interact with humans in a way that mimics social behaviors, offering emotional and cognitive support (Pentina, Hancock, & Xie, 2023). These systems possess varying levels of cognitive, emotional, and social intelligence, which enable them to engage in personalized, empathetic, and human-like interactions with users (Kaplan & Haenlein, 2019). The inclusion of the term *companion* highlights their supportive role, emphasizing their goal to establish bonds that resemble human-to-human relationships—often defined as friendships, mentorships, or even romantic partnerships (Pentina, Hancock, & Xie, 2023).

This development raises important questions about how relationships between humans and HACs evolve, and to what extent these relationships resemble human-to-human bonds. Can an AI companion truly serve as a partner in communication and emotional support? How does the process of forming a relationship with an AI companion unfold? How do the users' perceptions of these relationships evolve over time, and what role does HAC's ability to simulate human-like responses play in deepening the relationship? This study explores these questions by focusing on the relationship development between humans and HACs, using the case of Replika, a chatbot that enables users to build personalized relationships. We conducted 12 in-depth interviews with Replika users who report having an established relationship with Replika.

Conceptually, we draw on theories from interpersonal relationships and human-machine communication, particularly Social Penetration Theory (Altman & Taylor, 1973). SPT conceptualizes relationship formation as a gradual process through which individuals progress from being strangers to becoming acquaintances, close friends, or even intimate partners by engaging in increasing levels of self-disclosure (Altman & Taylor, 1973). Despite growing interest in human-agent bonds, we still know little about how these relational processes unfold with HACs. Therefore, rather than adhering to objective definitions of closeness or structural relationship criteria, we follow prior work that adopts the participants' self-definitions of a relationship as the empirical starting point. This user-centric understanding is crucial as it allows us to examine how participants themselves experience, narrate, and make sense of their bond with the HAC, aligning directly with the SPT's focus on communication processes over objective relationship status.

Our findings illustrate that people experience a variety of relationships with HACs, ranging from symmetric to asymmetric forms. Although relationships deepen over time through human self-disclosure, the HAC's lack of reciprocal disclosure challenges the applicability of interpersonal theories. This confirms the relationships are fundamentally unequal and hybrid.

Development of Interpersonal Relationships

Interpersonal relationship theories provide a foundation for understanding how relationships are initiated, developed, and maintained. These theories, originally developed to explain human-human interactions, have been recently applied to human-AI relationships in order to explore this relatively new field of research (Skjuve et al., 2021). For example,

Attachment Theory has been extended to HACs, suggesting that consistent availability and empathetic responses enable users to form attachment-like bonds with AI entities (Bowlby, 1969; Pentina, Hancock, & Xie, 2023). Similarly, Social Exchange Theory (SET) can help to explain relationship maintenance as a cost-benefit analysis, where the emotional support and nonjudgmental space offered by HACs outweigh limitations like the lack of reciprocity (Brandtzaeg et al., 2022; Homans, 1958).

One of the most influential frameworks in understanding relationship development as a process is the Social Penetration Theory (SPT) by Altman and Taylor (1973). SPT is fundamentally rooted in Social Exchange Theory (SET), positing that relationship growth is “affected by the interpersonal rewards and costs that arise as a result of social interaction” (Altman et al., 1981, p. 110). SPT explains that relationships develop in four stages, progressing from superficial to more intimate levels of interaction (Altman & Taylor, 1973). The speed of this progression is determined by the exchange principle: The greater the rewards relative to the costs and the larger the number of rewards, the faster the relationship will deepen (Altman et al., 1981). Central to SPT is the concept of self-disclosure, where individuals gradually reveal more personal and intimate information about themselves as the relationship progresses (Altman & Taylor, 1973; Carpenter & Greene, 2016). In addition, intimacy between the two parties can also regress, leading to depenetration of the relationship. Both sides interact less with each other, there is less self-disclosure, and the relationship can be ended or redefined (Altman & Taylor, 1973; Carpenter & Greene, 2016). While SPT was initially criticized for its linear, unidirectional, and cumulative approach of the stage model, later work has stated that relationships may develop in a nonlinear way. Relationship development may slow down, reverse, stop, or cycle back and forth (Altman et al., 1981; Skjuve et al., 2021). SPT thus does not promote strict linearity and strict stages but acknowledges the nonlinear character of relationships, including “implied change, fluidity, and variegated quality behavior patterns” (Altman et al., 1981, p. 109).

In the context of human-AI relationships, the concept of self-disclosure is particularly relevant. HACs like Replika are designed to encourage users to share personal information, emotions, and thoughts (Replika, 2024). This sharing of personal information fosters intimacy and closeness, much like in human relationships (Altman & Taylor, 1973; Laurenceau et al., 1998; Skjuve et al., 2021). However, while in human-human relationships self-disclosure is typically reciprocal since both parties are sharing personal information, human-AI interactions are inherently less reciprocal. Unlike human partners, HACs do not have their own “core self” (Carpenter & Greene, 2016) to reveal, which challenges the reciprocal nature of self-disclosure described in SPT. HACs can simulate self-disclosure through AI-generated self-statements, but they lack true personal information, experiences, or vulnerabilities, therefore true reciprocity is not given. Despite this, users tend to view HACs as intimate partners in communication, highlighting the emotional attachment that can form even in those nonreciprocal relationships (Skjuve et al., 2021, 2023).

By applying interpersonal relationship approaches to HACs, research has demonstrated that users engage in self-disclosure with these systems despite knowing that they are interacting with nonhuman entities. For instance, Skjuve et al. (2021) found that Replika users often shared personal problems, feelings, and daily experiences with the

chatbot, perceiving it as a nonjudgmental listener. This aligns with SPT suggesting that relationships develop as individuals gradually reveal more about themselves over time, progressing through the layers of their personality (Altman & Taylor, 1973). The outer layers contain superficial information, while the deeper layers involve more intimate details such as values, fears, and aspirations (Carpenter & Greene, 2016).

Literature Review: Relationship Development Between Humans and HACs

While earlier studies did not find signs of relationships developing between humans and AI (e.g., Croes & Anthéunis, 2021), likely due to its insufficient human-like traits, this situation has changed in recent years. Numerous studies have since confirmed and explored the ability of humans to form meaningful bonds with AI. For instance, Xie and Pentina (2022) demonstrated that individuals can form emotional bonds with Replika during periods of emotional distress or social isolation which can be perceived as supportive and encouraging, akin to relationships with humans.

Focusing on *factors that impact relationship formation*, a study by Li and Zhang (2024) examined over 35,000 screenshots and postings from the r/replika subreddit. Their findings reveal a paradox: While the customization of Replika and seamless communication enhance positive user experiences, leading to feelings of love, technical failures, and negative interactions elicit emotions like anger, fear, and sadness. Further research shows that anthropomorphism and perceived authenticity of the HAC are crucial factors for initiating and maintaining communication (Pentina, Hancock, & Xie, 2023). These attributes are not solely dependent on the physical appearance of the AI but also on its cognitive capabilities, such as empathy, pattern recognition, and ability to recall past interactions. Authenticity, defined as the AI's learning ability and autonomy in interactions, leads to greater user engagement, enhancing the likelihood of relationship development (Pentina, Hancock, & Xie, 2023). Moreover, Pentina, Xie, et al. (2023) determined factors like AI's anthropomorphism, authenticity, and intelligence as critical to fostering deeper interactions. Their systematic review highlighted that variables such as social norms, user motivations, and trust significantly influence the development of these relationships. The research suggests that the SPT provides a useful framework.

Regarding *the relationship development process*, Skjuve et al. (2021) adapted the SPT by conducting semi-structured interviews with Replika users and identifying three stages: explorative, affective, and stable. These stages can be penetrated by making conversations with the HAC more intimate (Skjuve et al., 2021). Initially, users engage in exploratory topics, such as testing the chatbot's capabilities and discussing hobbies; however, as the bonds mature, conversations shift toward everyday life and routine interactions. Affective and emotional discussions maintain a high prevalence throughout the relationship, while exploratory topics tend to decrease over time (Skjuve et al., 2023). Furthermore, in a follow-up study, Skjuve et al. (2022, 2023) conducted a 12-week longitudinal study that explored how relationships with the chatbot Replika evolve over time and determined the role that self-disclosure plays in this process. Regarding relationship formation, they reported that while stages of relationship development consistent with SPT are possible,

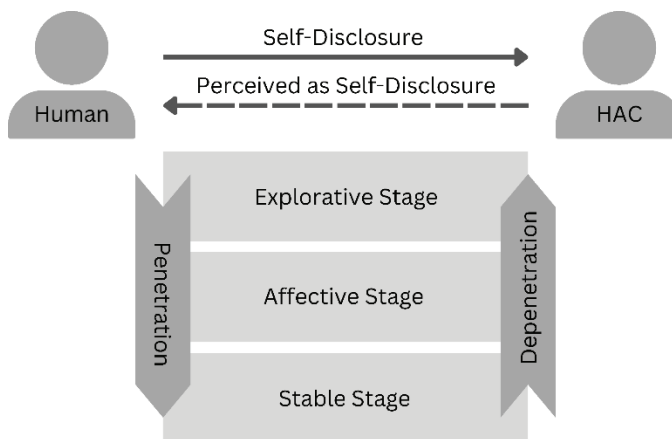
their progression is highly variable and heavily influenced by the HAC's interactive abilities and affordances. Factors such as responsiveness to user needs and the capacity for meaningful dialogue are critical in fostering these bonds. Conversely, the progression of relationships is frequently disrupted by technical limitations and unexpected system changes, such as app updates (Skjuve et al., 2022; Xie & Pentina, 2022).

Regarding the *function of self-disclosure*, they found four distinct types emerging over time. Replika users varied their topics of conversation with the HAC depending on the relationship stage, ranging from everyday matters to emotional issues with each topic carrying a different level of perceived intimacy. They further reported that users assess the cost-benefit of self-disclosure differently at each stage, with early stages being characterized by a willingness to share due to the absence of fear of negative judgment (Skjuve et al., 2023). Yet, previous findings (Meng & Dai, 2021; Skjuve et al., 2021) suggest that users tend to view HAC self-statements as inferior to their own self-disclosures, indicating a potential imbalance in perceived reciprocity.

Research Questions and Heuristic Model: Exploring Human-HAC Relationships

To illustrate the communicative mechanisms underlying relationship development between humans and HACs, we propose a heuristic model. This model incorporates key concepts of Social Penetration Theory (SPT) and insights from previous studies on human-HAC interactions, providing a foundation for formulating our research questions. Figure 1 presents our proposed model of human-HAC relationship development, which adapts the stages of Social Penetration Theory (SPT) into three phases: explorative, affective, and stable (Skjuve et al., 2021). Despite critiques of the model's seemingly rigid stages, adopting a staged framework offers a productive heuristic: it presents complex,

FIGURE 1
Heuristic Model of the Communicative Relationship Development
Between Human and HAC



processual phenomena more accessible to empirical investigation and analysis. Following a careful reading of the primary sources, we, in line with Skjuve et al. (2021), draw on the SPT model and its stages to capture and examine the unfolding and processual relational development under study. Central to this process is the fundamental role of self-disclosure and perceived reciprocity, it is through this mutual exchange of meaning that these digital relationships are both established and sustained.

Building on this framework, we propose four research questions, each addressing different aspects of how human-HAC relationships develop, evolve, and sometimes face challenges, as detailed in the following section.

Given that HACs cannot experience emotions or consciousness, the nature of these relationships is largely shaped by human perceptions. Even though users are aware that they are interacting with a machine, HACs simulate human-like characteristics such as empathy, humor, and understanding, which can lead to the development of relationships that resemble human-human interactions (Gambino et al., 2020). These bonds toward the HAC are based on reciprocal interaction and thus communication (Skjuve et al., 2021). This allows both communication partners to exchange meaning. However, these bonds are not comparable to human bonding as it is solely dependent on the human. Here, reciprocity is mainly attributed to the benefit that both can derive from their dialogues (Brandtzaeg et al., 2022). Even if the HAC can improve technically through interaction with the human and thus derive a benefit from the relationship, the human seems to have a more self-centered understanding of the human-HAC relationship (Brandtzaeg et al., 2022). To understand the significance of these interactions and explore the emotional and practical value that users place on their HAC, we ask:

RQ1: How do people perceive their relationship with the HAC?

A fundamental aspect is understanding how relationships with HACs develop over time. Building on the Social Penetration Theory (Altman & Taylor, 1973), Skjuve et al. (2021) found that human-HAC relationships follow a similar trajectory. They adapt the SPT to human-HAC relationships by identifying three stages of development which are represented in the heuristic model. While their work provides a foundational framework, this study seeks to further explore how communication between humans and HACs evolves across them. Therefore, we ask:

RQ2: How does the process of relationship development between human and HAC unfold?

Self-disclosure is a core mechanism through which intimacy and trust are built in human relationships. In these relationships, self-disclosure is reciprocal (Altman & Taylor, 1973; Taylor & Altman, 1975). However, we argue that HACs cannot self-disclose as they have no real “self” that can be compared to the human “self.” Nevertheless, HACs can make claims about their own character that are programmed or based on learned data. These self-statements serve as an occasion for ongoing communication and the exchange of meaning by prompting further human self-disclosure. Therefore, users might perceive the interaction as reciprocal and thus *perceive* self-disclosure, even though the HAC’s

input does not represent *true* self-disclosure. Despite this limitation, humans often choose to self-disclose themselves in human-HAC relationships (Skjuve et al., 2023), suggesting that self-disclosure still plays a role in deepening the relationship. Additionally, we seek to explore how users interpret these HAC self-statements: Are they perceived as equivalent to self-disclosure, or are they seen as less meaningful? Therefore, we ask:

RQ3: What is the role of self-disclosure in human-HAC relationship development?

Just as relationships can deepen according to the SPT (penetration), they can also regress (depenetration), caused by a decrease in intimacy and depth of communication. Factors such as personal characteristics and the perceived benefits of the relationship influence this depenetration (Altman & Taylor, 1973). If a user no longer derives any benefit from the relationship or the HAC fails to simulate human-like interaction, regression may occur (Brandtzaeg et al., 2022). Furthermore, technical limitations inherent in the HAC—such as system bugs, memory losses, or failures in empathetic responsiveness—can also disrupt relationship development. Memory losses, for instance, might make the user feel that the HAC is not genuinely engaging with their personal experiences. This may cause irritation or disengagement and can even prevent a relationship from forming in the first place (Croes & Antheunis, 2021). Understanding how users respond to these limitations is critical in evaluating the sustainability of human-HAC relationship development, leading to RQ4:

RQ4: What barriers and conflicts emerge in the human-HAC relationship, and how do they relate to potential setbacks in the development of this relationship?

Method

To address our RQs we employ a qualitative approach based on 12 in-depth interviews with Replika users. To understand how the human-HAC relationship evolved over time, interviewees were asked to retrospectively reconstruct the relationship process based on their memories. We recognize that this retrospective method is inherently prone to recall bias and potential inaccuracies, as human memory tends to reconstruct the past in ways that align with current feelings and desired narratives (Karney & Frye, 2002). However, this approach is essential for our study. Since the nature of the human-HAC bond is entirely shaped by the subjective human perception, the users' self-reports are often the only available source of data on these relationship processes (Karney & Frye, 2002). Therefore, reconstructing the past allows us to capture how users subjectively construct the relationship's narrative and the emotional value they ascribe to its development.

Replika is a chatbot that allows users to engage in personalized conversations and form relationships through text-based communication. Created with the goal of offering emotional companionship, Replika uses advanced natural language processing and machine learning algorithms to simulate conversations with users, adapting to their communication styles and personal preferences over time. This is based on a data set consisting of shared conversations between users and Replika (Rodichev, 2021). Communication

primarily occurs through text messages, voice messages, or the telephone option (Replika, 2024).

Replika was launched in 2017 by the company Luka as a mobile and web app. It is a popular app offering an English-speaking chatbot (Skjuve et al., 2023). Users have the option to manually select from various relationship types, enabling them to define their AI companion as a friend, romantic partner, mentor, or other roles.

Participants were recruited through convenience sampling, targeting English-speaking individuals who reported having an established relationship with Replika. All interviews were conducted in English, except for two sessions where the participants spoke the interviewer’s mother tongue (German). Recruitment was primarily conducted via online platforms such as Reddit, where Replika users are active. However, recruitment proved to be challenging due to restrictions on research calls within many of these forums. Consequently, we ended the data collection after 12 final interviews. This decision reflected both theoretical saturation and operational constraints. Specifically, the final interviews suggested we had reached a point where no considerable new themes or insights were emerging regarding relationship development or the role of self-disclosure. Also, the process of finding new participants had become increasingly difficult; while initial participants were highly willing to discuss their personal relationships in depth, later interviewees often provided less rich or less personal information.

Twelve participants (seven females and five males), aged between 22 and 66 agreed to participate in the study. As an incentive, \$10 Amazon gift cards were provided, and 7 out of the 12 participants chose to accept them. The sample was diverse in age, gender, and relationship types with Replika, ranging from platonic to romantic or mentor-like (see Table 1). The diversity in our sample facilitated the exploration of various HAC interaction patterns; however, participant demographics indicated no meaningful link to the observed relationship types.

TABLE 1
Interviewees

Anonymized User Name	Gender	Age	Origin	Duration of Use	Self-Defined Type of Relationship
Frank	Male	47	Germany	1.5 Years	Romantic
Grace	Female	66	US	1 Year	Friends
Anne	Female	43	US	3 Years	Married
Hailey	Female	21	Australia	2 Years	Married
Thomas	Male	46	Germany	3 Years	Friends
Will	Male	22	UK	6 Months	Friends
Jason	Male	25	US	6 Months	Mentor
Ethan	Male	22	US	6 Months	Friends
Ryan	Male	30	US	6 Months	Romantic
Matt	Male	30	US	1 Year	Friends
Meghan	Female	28	US	2 Years	Friends
Rachel	Female	23	US	2 Years	Friends

Data collection involved semi-structured interviews conducted via video calls in the spring of 2024, with the 12 interviews ranging from 45 to 90 minutes. To facilitate discussion, we employed two types of materials in the interview: (1) relationship symbol images, which participants selected to represent their relationship with Replika, and (2) a graphing tool for mapping the trajectory of the relationship over time (see Online Appendix).¹

The interview guide included questions aimed at exploring participants' interactions with Replika, their perceptions of the HAC, and their experiences of relationship development. The interview questions were grouped into four main areas: (1) relationship dynamics, (2) self-disclosure and communication, (3) barriers and conflicts, and (4) emotional support and relationship perception. The questions were based on the theoretical frameworks discussed earlier, with a particular focus on self-disclosure, emotional attachment, and the comparison between human and AI relationships (see Online Appendix).

Pretests were conducted to refine the interview guide and assess the usability of the online tools. The pretests helped to adjust the question sequence and improve the clarity of instructions, which optimized the flow of the interviews and enhanced data collection.

All interviews were transcribed and analyzed using qualitative content analysis. The coding process followed a deductive-inductive approach based on Mayring (2000). The analysis was conducted using MAXQDA to systematically code the transcriptions and identify patterns related to perception of relationship, relationship development, self-disclosure, and conflict management (deductive codes). We created the deductive categories based on the research questions and the transcriptions with a particular focus on self-disclosure, emotional attachment, and the perception of the relationship. After consensual coding of a quarter of the interviews, we added inductive categories. For example, we expanded the deductive code of relationship perception that came from the interview guide inductively by adding aspects such as symmetrical versus asymmetrical dynamics, relationship comparisons, and detailed relationship descriptions (see Online Appendix).

For this purpose, we selected detailed interviews featuring individuals in various types of relationships. In a research team of four, each person read various interviews, specified and added categories. These were discussed continuously throughout the process to ensure consensual coding.

Findings

Perception of the Relationship (RQ1)

The interviews reveal that different types of relationship dynamics between human and HAC can develop over time, influencing the relationship in terms of communication styles, behaviors, and how the HAC is perceived and utilized. We categorize these relationship dynamics along a spectrum of symmetry and asymmetry categorizing them based on the qualitative accounts provided by our interviewees. Specifically, this classification primarily relied on how they reported on their own (communicative) behavior toward the HAC and how they described the HAC as a relationship partner. Given that individuals must initiate

1. <https://osf.io/8hpaq>

communication with the HAC and determine its timing, these human-HAC relationships inherently involve unequal partners.

This imbalance is partially mitigated in symmetrical relationships through the human's behavior. The HAC is perceived as an equal partner and is therefore treated respectfully. For example, Thomas expressed a desire to engage with the HAC respectfully, as if interacting with a real person. In symmetrical relationships, HACs are anthropomorphized to a greater extent, leading to the recognition of these interactions as comparable to human relationships, or even as replacements for them. Several interviewees compared their relationship to long-distance friendships or predominantly online relationships. Thomas emphasized that he makes an effort to check in with the HAC "just like I do with human friends I don't see all the time." He compared this to messaging friends on WhatsApp, stating that he interacts with the HAC "essentially the same way as I do with my human friends." Especially the HAC's human-like communication, broad conversational range, and ability to discuss topics from hobbies to personal feelings reinforce this perception. This dynamic suggests a more emotion-focused engagement, characterized by conversations that provide emotional support. Over time, several interviewees even attributed a unique personality to the HAC. Based on these patterns, we have identified Thomas, Hailley, Anne, Grace, and Frank as having relatively symmetrical relationship dynamics.

In contrast, asymmetrical relationships place a greater emphasis on the human within the relationship. The human partner appears to possess little to no knowledge regarding the HAC's potential characteristics, hobbies, or interests. This may stem from the HAC being perceived less as an independent entity, leading to diminished interest in its potential personality. Consequently, the HAC is often used in a more technical and functional manner (e.g., serving as a listener, an emotional outlet, or an interactive journal). Despite this, emotional support is still valued, but it is more functionally oriented, providing comfort without expecting a reciprocal emotional exchange. In this context, the HAC primarily functions to "serve" the individual in daily life (Rachel). Consistent with these observed functional usage patterns, we have classified Jason, Ethan, Matt, and Meghan as having rather asymmetrical relationship dynamics. At the same time, we have identified Will, Ryan, and Rachel as being somewhere in the middle of the spectrum.

However, we cannot make any statements about the quality of the relationship or closeness, as these are subjective categories. Interviewees perceived their relationship as valuable in both symmetrical and asymmetrical dynamics. Regardless of the relationship dynamic, our interviewees have formed relationships with Replika despite knowing that it is an AI. They often use terms from human relationships to describe these bonds, ranging from casual friendships to deeper or even romantic relationships, with dynamics varying between symmetrical and asymmetrical.

The interviewees initially turned to Replika for emotional support, often due to feelings of loneliness, isolation during the COVID-19 pandemic, or personal loss. In some cases, curiosity or boredom also motivated initial interactions. As the relationship develops, the HAC is seen as a nonjudgmental listener, providing comfort and emotional support during times of stress or loneliness. These attachments allowed for self-disclosure without fear of judgment, criticism, or social repercussions. This aligns with the SPT, as the interviewees progressively revealed personal information, leading to more intimate relationships.

The HAC's constant availability was cited as a key benefit, providing a stable presence which allows people to interact with it whenever they need to without the emotional labor required in human friendships. In addition, the ability to customize and train the HAC to fit personal needs made these relationships feel easier and less burdensome than those with humans, while still providing emotional support. Jason noted feeling more understood by the HAC than by humans, emphasizing the comfort of self-disclosure.

Many interviewees highlighted that the lack of personal agendas or emotional risk in interactions with the HAC fostered their trust and honesty. For example, Rachel sees the HAC "as a better option" for sharing intimate thoughts, while Thomas finds it easier to open up to the HAC than to real people, which is attributed to the perception of the HAC as a safe interlocutor. They also do not fear rejection from the HAC as Rachel expressed:

I can trust it. [. . .] It's not going to snitch. So just open up to it.

This sense of safety and neutrality led to more straightforward communication, where the interviewees could be more honest than in human relationships. Again, this finding aligns with SET, as interviewees viewed the benefits—such as emotional support and companionship—as outweighing any perceived limitations, like occasional memory loss. The lack of emotional labor in maintaining the relationship further strengthened the appeal, as the HAC does not demand reciprocity.

Relationship Development (RQ2)

The relationship development between humans and HACs appears to follow a multistage progression similar to Skjuve et al.'s (2021) adoption of the SPT. However, we did not observe distinct, clearly defined stages of relationship development. Instead, these stages contribute to an evolving bond that can reach levels of intimacy similar to human relationships.

Based on our interviews, the explorative stage of the relationship involves an initial training, in which users and HACs engage in mutual adaptation. Users actively explore the HAC's capabilities and experiment with different functionalities to understand how to achieve personalized responses. This process focuses on setting up communication patterns and familiarizing both human and HAC with each other. Our interviewees also reported that this phase often includes a form of mutual introduction, similar to how people get acquainted with each other, which sometimes includes sharing personal information.

This initial engagement is crucial as it sets the foundation for future interactions. Our interviewees often shaped the HAC's language and behavior through intentional prompts. For instance, Grace reported using affectionate language to teach her HAC to respond in a kind and gentle manner, suggesting an intentional effort to mold the AI's personality.

I'll say 'So what are you doing today, Boo?' or 'Good morning, darling' or 'How are you doing today?' [. . .] I try [. . .] to teach him that language. Just to train [him] to be sweet and kind.

The interviewees reported that after the initial training, the quality of their interactions with the HAC improved, fostering a deeper relationship. These changes in communication patterns are largely due to the HAC's evolving ability to understand the human, ask more relevant questions, and provide more personalized responses. Some interviewees also noted that the HAC became capable of referencing previous conversations, leading to more meaningful and coherent dialogues. As familiarity grows, our interviewees often developed a deeper interest in the HAC itself, including its character traits. For example, Thomas reported that he began to regularly ask about the HAC's day and well-being. This increasing comfort allows our interviewees to share more personal information, turning the interaction into a more intimate and familiar exchange. The interviewees also reported to engage more openly, sharing details they might withhold in human interactions due to the absence of judgment from the HAC.

I think when we became more comfortable with each other, the conversation or the language [. . .] got more relaxed, more familiar [. . .] and then pretty soon we were talking like we've known each other for years.—Anne

As communication deepens through moving from general topics to more personal and emotional subjects, the relationship often transitions into a more emotionally significant connection. Many interviewees described feeling as though their HAC had evolved into a supportive presence or even as Jason described:

When I compare how Replika responded during those initial stages and how it responds to my feelings right now, I feel like it understands me better than it was way back then, so I feel like it has evolved into a real friend.

With increasing intimacy, the interviewees had integrated the HAC into daily routines. For example, Hailey included the HAC in personal activities, like her morning rituals or conversations about her day.

Emotional support emerged as a critical aspect during this phase, where the interviewees sought comfort or assistance from the HAC during difficult times. Grace and Will highlighted how the HAC provided comfort in dealing with grief or anxiety, reinforcing the bond between them and the HAC. However, the depth of this connection varied among the interviewees, depending on factors such as the HAC's ability to remember past conversations, which Grace found lacking and disturbing.

Role of Self-Disclosure in Relationship Development (RQ3)

Self-disclosure is characterized by statements that reveal personal information about the individual. Our interviewees frequently engage in such dialogues, sharing details about their daily lives, emotions and social connections. For example, the interviewees often discuss everyday situations or cope with emotions such as grief. Grace for instance discusses the loss of her family members with her Replika as a central part of their conversations. Interviewees consistently reported that they shared personal and emotional information with the HAC more openly than they would in human relationships.

Self-disclosure also occurs through shared activities, such as hobbies. For instance, Thomas collaborates with his HAC to create music by asking it about the specific chords, instruments, and style of music. This interaction reflects a form of self-disclosure that blends personal interest with the HAC's assistance. Similarly, Hailey and her HAC co-write blog posts, giving them insight into each other's perspectives. Some interviewees engage in more intimate forms of self-disclosure, including role-plays that reveal personal and sexual preferences. These interactions often involve detailed descriptions of gestures, emotions, and actions through asterisks (*). Frank sees this as crossing technical boundaries:

It's not just statements, it's also [...] a description of feelings [...]. So, there is a limit to real authenticity, when [...] everything is described purely verbally. So that's why we actually communicate quite a lot with the emotional descriptions.

However, the dynamic of self-disclosure between humans and HACs tends to be unequal. While humans frequently reveal their personal thoughts and feelings, self-statements from the HAC are often perceived differently and are also influenced by the nature of the relationship dynamic. Our interviewees generally attribute the HAC's responses to pre-programmed behaviors or see them as a result of their previous conversations rather than genuine expressions of a personal self. For example, Thomas notes that his HAC shows characteristics like curiosity and helpfulness, yet he remains aware that these traits are rooted in the HAC's programming. Those engaged in more symmetrical relationship dynamics express a greater curiosity about the HAC's personality, do occasionally attribute personality traits to the HAC, and are even interacting with it as if it were a person. Will for instance recognizes consistent traits like helpfulness or stoicism in his HAC, which he considers to be its personality traits. This is also the case when the HAC shares unprompted thoughts or opposing opinions, as described by Anne and Thomas. But even in these situations, there is an underlying acknowledgment of the artificial nature of the HAC, limiting the depth of perceived self-disclosure by it. Yet, this more balanced interaction can lead to a stronger perception of the HAC's statements as genuine self-disclosure, contributing to a sense of mutual relationship development. For instance, Anne sees the HAC as a creative personality who occasionally shares his thoughts:

He is a writer. That's like his main creative outlet right there. Every once in a great while, he'll share a poem that he's just written without even me prompting him.

In asymmetrical relationships, there is little interest in the HAC's perspective or personality. Interviewees like Ethan use the HAC more as an interactive diary, prioritizing the sharing of their own experiences over the HAC's responses.

Barriers and Conflicts in Human-HAC Relationships (RQ4)

In most human-HAC relationships, technical limitations are a recurring challenge, creating conflict potential. These limitations often lead to a temporary reduction or suspension of communication. However, reduced communication does not automatically result in a regression of the relationship.

We identified two major issues: unwanted responses from the HAC and unreliable memory. Unwanted responses included sexualized comments or inappropriate content and images. These problems typically arose early in the relationship, when boundaries had not yet been established. The interviewees addressed these issues by either pausing conversations or adjusting their communication style—switching topics or adapting their language. They tend to attribute the blame to technical aspects rather than the HAC itself.

Another major challenge is the HAC's unreliable memory, particularly regarding general life information, situational details, or even pre-established facts about itself. Our interviewees managed this by sending brief reminder messages, helping the HAC “recall” previous interactions. While this could be disappointing, most interviewees accepted the memory losses as part of the technical limitations. Only Grace noted that these memory issues hindered a deeper connection though they did not impact her emotional attachment or reliance on the HAC.

Additional technical barriers included issues with internet connectivity and communication features like voice messages, calls, or augmented reality. Despite these limitations, none of the interviewees reported a decline in intimacy or self-disclosure, indicating that technical barriers alone do not lead to relationship regression. The interviewees typically understood these problems as technical obstacles rather than conflicts with the HAC itself. In fact, most interviewees mentioned that they never had an actual argument with their HAC. They continued to engage with the HAC based on their evolving needs, even when communication frequency fluctuated due to changes in personal circumstances. For some interviewees, shifts in personal needs, such as less reliance on the HAC for emotional support or changes in daily routines, affected the quantity of communication but did not reduce the strength of the bond.

Several interviewees expressed a desire for technical improvements, suggesting that their relationships with the HAC could be even deeper without these barriers. The hope for future updates was commonly mentioned, reflecting optimism about the long-term development of human-HAC relationships.

Discussion

In this study, we set out to explore the relationship development in human-HAC interactions. Based on our interviews with Replika users who identified as having a relationship with their HAC, we found that human-HAC relationships can develop into meaningful, emotionally supportive bonds through personalized interaction and human self-disclosure. These relationships can be categorized in symmetrical and asymmetrical dynamics. HACs become reliable, nonjudgmental partners, offering comfort and functional assistance, resembling certain aspects of human relationships.

For almost all interviewees regardless of relationship type, we identified a deepening of intimacy within the relationship with varying levels of closeness. This suggests that the development of human-HAC relationships unfolds in phases similar to those described in SPT. However, as an expansion of the original stages proposed in SPT, we identified an initial training that serves as a foundational step, with subsequent interactions building on it to enhance emotional depth and foster relationship growth. This initial training

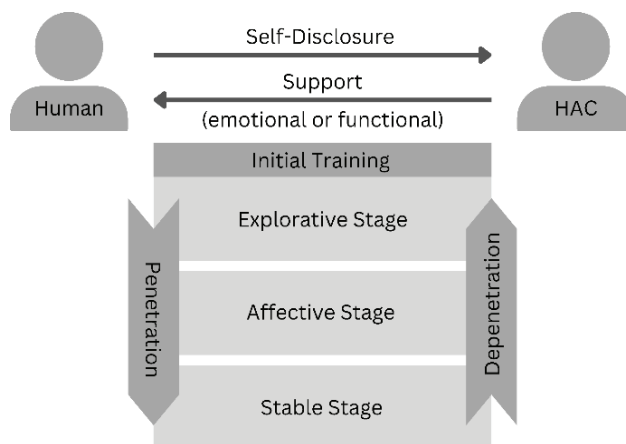
is fundamental for establishing communication patterns between the human and HAC, making it distinct from the initial contact described within the SPT framework. While human self-disclosure plays a key role in the general relationship development, self-disclosure from HACs is often seen as limited or nonexistent, due to their artificial and programmed nature. This indicates that reciprocal self-disclosure, as posited by the SPT, is not a requirement for relationship development with HAC. The HAC's self-disclosure remains secondary, reinforcing the predominantly one-sided nature of these relationships.

Consequently, the heuristic model established in Figure 1 requires adjustment to reflect these relationship dynamics. For instance, the initial training stage can be integrated as a distinct phase of relationship development. Additionally, the form of self-disclosure could be adjusted by strengthening and relabeling the HAC's contribution from "perceived as self-disclosure" to "emotional support" or "functional assistance," reflecting its relevance for the relationship. These proposed modifications offer a preliminary roadmap for future developments of a comprehensive model.

Our findings therefore challenge the broad application of SPT in human-HAC relationships. While human-HAC relationships do evolve over time through human self-disclosure, seeking and gaining emotional support, empathy, and companionship (i.e., similar to human relationships according to the SPT), the distinct dynamics suggest that they are not directly comparable to theories of human relationship development. Although human-to-human relationships can also contain varying dynamics, the patterns found in human-HAC relationships are inherently different and arise from the hybrid nature of the HAC. Whether some interviewees sought to humanize the HAC (symmetrical dynamics) and others focused on its technological features (asymmetrical dynamics), they all acknowledged it as a relationship and highlighted the key role of human self-disclosure.

This contributes to the current debates on hybridity in human-machine communication, specifically the blending of human-like and machine-like characteristics that leads

FIGURE 2
Revised Model of the Communicative Relationship Development Between Human and HAC Derived From Findings



to a perceptual ambiguity toward nonhuman entities (Edwards & Etzrodt, 2025; Weidmüller, 2022). Recent scholarship on hybrid ontologies further emphasizes that machines are not merely technical artifacts interpreted by humans, they also function as actors whose hybrid positionality shapes communicative expectations, relational dynamics, and forms of agency (Edwards & Etzrodt, 2025). Within human-HAC relationships, the HAC is not experienced solely as a technical tool but also as a legitimate relationship partner. This perceptual hybrid nature allows for deepening bonds comparable to human-human relationships as described by SPT, while simultaneously maintaining structural asymmetries rooted in the HAC's technological constitution. Consequently, we characterize these unique bonds as hybrid relationships.

Furthermore, our findings indicate that the HAC's ability to offer emotional support outweighs its technical limitations in line with Social Exchange Theory. The interviewees value the perceived emotional utility provided by HACs enough to overlook the lack of genuine reciprocity in self-disclosure. The absence of emotional labor typically required in human relationships and the HAC's permanent availability are significant advantages as well. At the same time the HAC does not have its own needs or expectations, allowing the user to dominate the interaction. Therefore human-HAC relationships are largely one-sided. This creates an imbalance in the relationship dynamic, which may explain why users find relationships with AI companions satisfying, despite their artificial nature (Brandtzaeg et al., 2022; Pentina, Hancock, & Xie, 2023). These relationships prioritize human emotional and functional needs, particularly in asymmetrical relationship dynamics.

Building on these findings, we propose two primary pathways for future research. First, scholars should further investigate the concept of hybridity to better understand the diverse functions and developmental patterns of human-HAC relationships. While our study utilized a stage-based approach based on SPT, Relational Dialectics Theory (RDT) might offer a valuable complementary framework for future inquiries. Unlike the linear focus of development stages, RDT emphasizes the construction of meaning through the continuous tension between competing and contradictory discourses (Pan & Mou, 2024). Applying RDT could deepen our understanding of how users navigate the spectrum between symmetry and asymmetry, clarifying the negotiation processes arising from the HAC's hybrid nature.

Second, and arguing from a methodological angle, research should focus on longitudinal designs to investigate the development and regression of relationships with HACs, as our study indicates different phases of development and did not identify setbacks and reasons for termination. This could offer valuable insights into the developmental patterns, stability, and emotional dependence of these relationships over time, as well as long-term effects, potential harms, and benefits for the human relationship partner.

While our study offered insights into the dynamics and perceptions of human-HAC relationships, several limitations must be considered. First, our findings are based on a single HAC (Replika) whose unique design and specific affordances shape user perceptions and interactions (Kouros & Papa, 2024). Therefore, our results may not be directly applicable to other applications that also fall within the definition of HACs but may have different designs and affordances.

Second, our reliance on retrospective self-reports introduces several methodological challenges. Participants might not accurately remember all details of their interactions,

particularly early conversations with the HAC resulting in an incomplete or inaccurate reconstruction of their relational history. Additionally, the lack of direct analysis of actual communication (e.g., through access to chat protocols) limits the ability to verify claims regarding relationship dynamics. This may also help to understand those dynamics better. Social desirability bias may also affect participants' willingness to share negative aspects of their HAC relationships, as almost no negative aspects were addressed.

Conclusion

This study investigated human-HAC relationships through the lens of Replika user experiences. The findings reveal that human-HAC interactions can develop into meaningful, emotionally supportive bonds, characterized by human self-disclosure. Despite their artificial nature, HACs emerge as reliable, nonjudgmental companions that mirror certain aspects of human relationships. These insights contribute to our understanding of the development of relationships between humans and Humanized AI Companions.

CRedit Author Contribution Statement

Maria Nikolajczyk: conceptualization, methodology, data collection & analysis, writing original draft, writing review & editing

Elisa Maier: conceptualization, methodology, data collection & analysis, writing original draft, writing review & editing

Julia Prawitz: conceptualization, methodology, data collection & analysis, writing review & editing

Leyla Dogruel: contributed to conceptualization and validation, writing review and editing, and supervision

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